

USCC Eastern Cyber Camp

Study Quiz & Flashcard Guide

Self-Assessment Workbook | Days 1-5

Prepared for: Brandonn

"Test your knowledge. Track your progress. Earn every flag."

HOW TO USE THIS GUIDE

This workbook is your personal self-assessment tool for the full USCC Eastern Cyber Camp curriculum. It covers all five days of training: Linux Forensics, Memory Forensics, Disk Forensics, Mobile & Malware Analysis, and Web / Bug Hunting. Work through each section actively and honestly — the goal is real mastery, not just checking boxes.

► Study Method

1. Work through each section **without looking at the answers first** — cover the answer column or key with a sheet of paper.
2. Use the Answer Key at the end of each section to self-grade immediately after completing it.
3. Mark any missed questions, then **re-attempt them after 24 hours** from memory.
4. Log your scores in the **Progress Tracker** at the back of this guide.

Scoring Scale: 90-100% = **MASTERY** | 70-89% = **REVIEW** | Below 70% =

RE-STUDY

SECTION 1 - LINUX FORENSICS | DAY 1

Part A — Command Flashcards (20 Cards)

Cover the "Answer" column. Recall the answer aloud before revealing.

Prompt / Question	Answer
1. What command lists all running processes with user and resource info?	ps aux
2. What command shows open network ports ss -tulnp OR netstat -tulnp	

Prompt / Question	Answer
and listening services?	
3. Where is the bash command history stored?	<code>~/.bash_history</code>
4. What command finds files modified in the last 24 hours?	<code>find / -mtime -1</code>
5. What command identifies a file's true type by magic bytes?	<code>file <filename></code>
6. What log file contains SSH login attempts on Debian/Ubuntu?	<code>/var/log/auth.log</code>
7. What command extracts human-readable strings from a binary?	<code>strings <filename></code>
8. What command lists all user accounts on a Linux system?	<code>cat /etc/passwd</code>
9. What command shows the current user's cron jobs?	<code>crontab -l</code>
10. What command shows who is currently logged in?	<code>w</code> OR <code>who</code>
11. What does the <code>-mtime -1</code> flag mean in a <code>find</code> command?	Files modified within the last 24 hours (1 day)
12. Where are system cron jobs stored?	<code>/etc/crontab</code> and <code>/etc/cron.*</code>
13. What command shows the access, modify, and change times of a file?	<code>stat <filename></code>
14. What command displays a hex dump of a file?	<code>xxd <filename></code>
15. What log contains kernel-level events?	<code>/var/log/kern.log</code>
16. What command lists all systemd services?	<code>systemctl list-units --type=service</code>
17. What directories are common attacker staging areas?	<code>/tmp</code> and <code>/dev/shm</code>
18. What command shows the current user's environment variables?	<code>env</code>

Prompt / Question	Answer
19. What does SUID mean and why is it suspicious?	Set User ID – runs file with owner's privileges; can be exploited for privilege escalation
20. What command shows the login history of all users?	last

Part B — Multiple Choice Quiz (10 Questions)

Circle or note your answer before checking the answer key below.

Q1. Which file contains hashed passwords on a Linux system?

1. /etc/passwd
2. /etc/shadow
3. /etc/group
4. /etc/hosts

Q2. An attacker wants to persist after reboot. Which location would they most likely modify?

1. /var/log/syslog
2. /etc/shadow
3. /etc/crontab
4. /proc/meminfo

Q3. You find a process named "sshd" making outbound connections to a foreign IP. What is the FIRST thing you should do?

1. Kill the process
2. Reboot the system
3. Document the PID and connection details
4. Delete the binary

Q4. The command `find / -perm /4000` searches for:

1. World-writable files
2. SUID binaries
3. Files modified today
4. Hidden files

Q5. Which command would you use to see BOTH screen output AND save it to a

file simultaneously?

1. redirect
2. tee
3. cat
4. more

Q6. Where are systemd service unit files typically located?

1. /etc/init.d/
2. /usr/share/services/
3. /etc/systemd/system/
4. /var/run/

Q7. The last command reads from which log file?

1. /var/log/auth.log
2. /var/log/wtmp
3. /var/log/syslog
4. /etc/passwd

Q8. Which of the following best describes the *chain of custody*?

1. The order in which commands are run
2. Documentation of who handled evidence and when
3. The sequence of attacker actions
4. A list of running processes

Q9. You want to search for the string "password" in all files in /home. Which command works?

1. find /home -name "password"
2. grep -r "password" /home
3. strings /home
4. cat /home | grep password

Q10. What is *volatile data*?

1. Data stored on disk
2. Data that persists after shutdown

3. Data that is lost when power is removed

4. Encrypted data

► SECTION 1 - PART B ANSWER KEY

Q#	Answer	Explanation
Q1	B	/etc/shadow stores hashed passwords. /etc/passwd holds account info only — not hashes on modern systems.
Q2	C	/etc/crontab entries survive reboots and execute commands on a schedule — a classic persistence location.
Q3	C	Always document before acting . Killing the process first destroys volatile data (open file handles, network state, memory contents).
Q4	B	/4000 is the SUID bit mask. SUID binaries run with the file owner's privileges — a major escalation risk.
Q5	B	<code>command tee output.txt</code> prints to screen and saves to file simultaneously. Essential for forensic logging.
Q6	C	User-installed and custom systemd service unit files live in /etc/systemd/system/.
Q7	B	last reads from /var/log/wtmp, a binary login database. Attackers sometimes wipe this file.
Q8	B	Chain of custody documents every person who accessed/handled evidence to ensure legal admissibility in court.
Q9	B	<code>grep -r "password" /home</code> searches recursively through all files in the directory.
Q10	C	Volatile data (RAM, open connections, running processes) disappears the instant the machine is powered off.

SECTION 2 - MEMORY FORENSICS | DAY 2

Part A — Volatility 3 Command Flashcards (20 Cards)

Cover the "Plugin / Command" column. Recall the correct Volatility 3 command for each task.

Task	Volatility 3 Plugin / Command
1. List all running processes	<code>windows.pslist</code>
2. Show process tree (parent-child relationships)	<code>windows.pstree</code>
3. Find hidden / rootkit processes (compare with pslist)	<code>windows.psscan</code>
4. Show command-line arguments for each process	<code>windows.cmdline</code>
5. List DLLs loaded by each process	<code>windows.dlllist</code>
6. Show active and closed network connections	<code>windows.netstat</code>
7. Scan for all network artifacts in memory	<code>windows.netscan</code>
8. Find files referenced in memory	<code>windows.filescan</code>
9. Dump files from a specific process	<code>windows.dumpfiles --pid <PID></code>
10. List loaded registry hives	<code>windows.registry.hivelist</code>
11. Check Run registry keys for persistence	<code>windows.registry.printkey --key "SOFTWARE\Microsoft\Windows\CurrentVersion\Run"</code>
12. Find suspicious RWX (Read-Write-Execute) memory regions	<code>windows.malfind</code>
13. Dump password hashes from SAM	<code>windows.hashdump</code>
14. List Linux processes from a memory image	<code>linux.pslist</code>
15. Recover bash command history from RAM	<code>linux.bash</code>
16. Show Linux network connections from memory	<code>linux.netstat</code>
17. Find injected memory regions in a Linux image	<code>linux.malfind</code>
18. Show Virtual Address Descriptor info	<code>windows.vadinfo</code>
19. List process privileges	<code>windows.privileges</code>
20. Show kernel modules loaded in	<code>windows.modules</code>

memory

Part B — Fill in the Blank (10 Questions)

Complete each sentence from memory. Write your answers before checking the key below.

1. The Volatility plugin _____ compares process lists to find hidden processes.
2. Memory that is lost when power is cut is called _____ data.
3. A process of _____ involves injecting code into a legitimate process's memory space.
4. The plugin windows. _____ detects memory regions with Read-Write-Execute permissions.
5. An encoded PowerShell command uses the flag _____ (or -enc) on the command line.
6. DLLs loaded from _____ or _____ directories are highly suspicious.
7. To verify a memory image has not been tampered with, compute its _____.
8. The tool _____ is used to acquire RAM from a live Linux system.
9. svchost.exe should always be spawned by _____, not explorer.exe.
10. The Volatility plugin windows. _____ extracts LSA secrets from memory.

► SECTION 2 - PART B ANSWER KEY

Answer

- 1 psscan - Cross-references physical memory to find processes pslist cannot see (rootkit technique).
- 2 **Volatile** - RAM, CPU registers, ARP cache, open connections.
- 3 **Process injection / process hollowing** - Malware hides inside a trusted process's address space.
- 4 malfind - RWX regions are abnormal; legitimate code is typically read-execute only.
- 5 -EncodedCommand - Base64-encoded commands are a common obfuscation technique.
- 6 %TEMP% / %APPDATA% - Also watch C:\Users\Public.

- 7 **SHA256 hash** - Cryptographic hash ensures integrity for chain of custody.
- 8 **LiME** (Linux Memory Extractor) - Kernel module that dumps RAM to a file or network socket.
- 9 `services.exe` - If `svchost.exe` parent is anything else, it is likely malicious.
- 10 `lsadump` - Extracts cached credentials and LSA secrets from memory.
-

SECTION 3 - DISK FORENSICS | DAY 3

Part A — Windows Event ID Flashcards

Cover the "Meaning" column. Recall what each Event ID signifies.

Event ID	Meaning / Significance
4624	Successful logon to a Windows system. Note the logon type (2=interactive, 3=network, 10=remote interactive).
4625	Failed logon attempt. Multiple rapid failures = brute-force attack indicator.
4634	Account was logged off (session ended).
4648	Logon using explicit credentials (e.g., runas). Often seen in lateral movement / Pass-the-Hash attacks.
4688	A new process was created. Shows parent/child process name — key for tracking execution chains.
4698	A scheduled task was created. Attackers use scheduled tasks for persistence.
4720	A new user account was created. Unexpected creation = possible backdoor account.
4732	A member was added to a security-enabled local group (e.g., Administrators group).
7045	A new Windows service was installed. Attackers install services for persistence or privilege escalation.
1102	Security audit log was CLEARED. Critical red flag — attacker is covering tracks. Investigate immediately.
4768	Kerberos TGT (Ticket Granting Ticket) was requested. Watch for unusual accounts or

off-hours requests (Golden Ticket attacks).

4776 The domain controller attempted to validate credentials. NTLM credential validation — watch for repeated failures.

4663 An attempt was made to access an object (file/folder). Used to track who read/wrote sensitive files.

4656 A handle to an object was requested. Precedes 4663 — shows intent to access an object.

4657 A registry value was modified. Critical for detecting persistence via registry run keys.

Part B — Multiple Choice Quiz (8 Questions)

Q1. What does the `mm!ls` command do?

1. Lists running processes
2. Shows partition layout of a disk image
3. Mounts a disk image
4. Scans for malware

Q2. Which command creates a bit-for-bit forensic copy of a disk?

1. `cp -r`
2. `rsync`
3. `dd if=/dev/sdb of=disk.img`
4. `tar -czf`

Q3. Prefetch files (.pf) are significant because they:

1. Store user passwords
2. Prove a program was executed even if later deleted
3. Log network connections
4. Contain browser history

Q4. Event ID 1102 indicates:

1. A new user was created
2. A service was installed
3. The audit/security log was cleared

4. A failed logon

Q5. What is *file carving*?

1. Deleting files securely
2. Recovering files by magic bytes regardless of file system
3. Encrypting evidence
4. Listing file permissions

Q6. The Master File Table (MFT) belongs to which file system?

1. ext4
2. FAT32
3. NTFS
4. APFS

Q7. You should always mount a forensic image with which option?

1. -o rw
2. -o exec
3. -o ro (read-only)
4. -o sync

Q8. Which Sleuth Kit command recursively lists all files including deleted ones?

1. mmls
2. fsstat
3. fls -r
4. icat

► SECTION 3 - PART B ANSWER KEY

Q#	Answer	Explanation
Q1	B	mmls (from The Sleuth Kit) displays the partition layout — offsets, sizes, and types of partitions in a disk image.
Q2	C	dd if=/dev/sdb of=disk.img creates a raw sector-by-sector copy. Always hash before and after to verify integrity.
Q3	B	Windows Prefetch files record evidence of execution including timestamp and run count — even if the executable was deleted.
Q4	C	Event 1102 = Security log cleared. Critical indicator that an attacker is destroying

Q#	Answer	Explanation
		evidence. Correlate with Event 4624 to find who cleared it.
Q5	B	File carving searches raw data for magic bytes (file signatures like FF D8 FF for JPEG) to reconstruct files independent of the file system.
Q6	C	The MFT (Master File Table) is the core metadata structure of NTFS — every file and directory has at least one MFT entry.
Q7	C	Always mount forensic images read-only (-o ro) to prevent accidental modification of evidence.
Q8	C	<code>fls -r</code> recursively lists file names including deleted entries (marked with an asterisk *) — essential for finding deleted artifacts.

SECTION 4 - MOBILE & MALWARE ANALYSIS | DAY 4

Part A — Mobile Forensics Flashcards (15 Cards)

Cover the "Answer" column. Recall each answer before revealing.

Prompt / Question	Answer
1. What ADB command lists all connected Android devices?	<code>adb devices</code>
2. What ADB command pulls a file from an Android device to your computer?	<code>adb pull <device path> <local path></code>
3. What ADB command opens a remote shell on a connected Android device?	<code>adb shell</code>
4. Where is SMS data stored on Android?	<code>/data/data/com.android.providers.telephony/databases/mmssms.db</code>
5. Where is the call log database stored on Android?	<code>/data/data/com.android.providers.contacts/databases/contacts2.db</code>
6. What type of acquisition physically reads raw NAND flash chips?	Chip-off acquisition (most invasive; requires desoldering the chip)
7. What is a logical acquisition?	Acquiring only the files and folders the OS exposes —

Prompt / Question	Answer
	fastest but least complete method
8. What is a physical acquisition?	A bit-for-bit copy of the entire storage chip — captures deleted data and unallocated space
9. What command extracts GPS metadata from an image file?	<code>exiftool <filename></code>
10. What exiftool flag shows only GPS-related fields?	<code>exiftool -gps:all <filename></code>
11. Where are iTunes iOS backups stored on a Windows machine?	<code>C:\Users\<username>\AppData\Roaming\Apple Computer\MobileSync\Backup\</code>
12. What database format does Android use for most application data?	SQLite (.db files) — browse with <code>sqlite3</code> or DB Browser for SQLite
13. What ADB command backs up all app data from an Android device?	<code>adb backup -apk -all -f backup.ab</code>
14. What is the significance of the /data/data/ directory on Android?	Contains private application data for all installed apps — requires root or ADB to access
15. What command on Android shows installed packages?	<code>adb shell pm list packages</code>

Part B — Malware Analysis Scenario Questions (5 Scenarios)

Read each scenario. Write your response steps before reading the answer.

Scenario 1: Suspicious .EXE — Not Yet Executed

You receive a suspicious .exe file via email. You have NOT run it. What do you do?

► Answer - Step by Step

1 Hash it first: `sha256sum suspicious.exe` — document the hash for chain of

custody.

2 Submit the hash to **VirusTotal** — do NOT upload the file directly unless

authorized.

3 Run `file suspicious.exe` to confirm true file type.

4 Run `strings suspicious.exe` to extract readable text (IPs, URLs, registry keys,

commands).

5 Analyze PE headers with **PEStudio** — check imports, entropy, timestamps, and

sections.

6 Check imported APIs for suspicious functions: `VirtualAlloc`,

`WriteProcessMemory`, `CreateRemoteThread` (injection indicators).

7 Open in **Ghidra** for static disassembly if deeper analysis is needed.

Scenario 2: HTTP Beaconing Every 60 Seconds

You run a sample in a VM and observe it making HTTP requests every 60 seconds to the

same IP. What does this indicate and what do you document?

► **Answer**

This is almost certainly a **C2 (Command and Control) beacon**. Document all of

the following:

- Destination IP address and domain (if DNS resolved)

- Destination port and protocol (HTTP/HTTPS)

- HTTP method (GET vs POST) and User-Agent string

- Beacon interval (60 seconds — known as "jitter-free beaoning")

- Any data sent in the request body

Use **FakeNet-NG** to intercept and simulate C2 server responses to observe malware behavior.

Scenario 3: Registry Run Key Persistence

Malware adds a key to HKEY_CURRENT_USER\Software\Microsoft\Windows\

CurrentVersion\Run. What does this mean and what do you do?

► **Answer**

This is a **registry persistence mechanism**. The malware will re-execute on every

user login automatically. Steps:

1 Document the **full registry path, key name, and value** (the path to the malicious executable).

2 Use **Regshot** (take a snapshot before and after infection) to compare all registry changes.

3 Cross-reference with Windows Event ID **4657** (registry value modified).

4 Note: HKCU\Run only requires user-level privileges — no admin access needed for this persistence method.

Scenario 4: windows.malfind Shows RWX Region in explorer.exe

During memory analysis, windows.malfind identifies a memory region with RWX permissions inside explorer.exe. What does this indicate?

Answer

RWX (Read-Write-Execute) memory regions inside a trusted process are a **strong indicator of code injection or process hollowing**. Steps:

1 Note the PID of explorer.exe and the suspicious memory region's virtual address.

2 Dump the region: windows.dumpfiles --pid <PID> or dump directly with

`windows.vadump.`

3 Submit the dumped binary to **VirusTotal** for reputation check.

4 Analyze the dumped shellcode with **Ghidra** or **Cutter** for static analysis.

5. Look for signatures of common injectors (e.g., Metasploit stager, Cobalt Strike shellcode).

Scenario 5: Embedded Command String in Binary

Running strings on a binary reveals: `cmd.exe /c net user admin P@ssw0rd /add.`

What is happening and what do you do?

► **Answer**

This command adds a local administrator account — a classic **persistence and**

privilege escalation technique. Steps:

1 **Document the exact string verbatim** — exact syntax matters for evidence.

2 Note the username (admin) and password (P@ssw0rd) — check if this account

exists: `net user.`

3 If in dynamic analysis, check **ProcMon** (Process Monitor) to confirm whether the

command actually executed.

4. Check Windows Event ID **4720** (new user created) and **4732** (user added to admin

group) in the Security log.

5. If on a live system, disable and remove the rogue account immediately after

documentation.

SECTION 5 - WEB / BUG HUNTING | DAY 5

Part A — OWASP Top 10 Flashcards

Cover the right column. State the description and an example before revealing.

Vulnerability	One-Line Description + Example
A01 - Broken Access Control	Users can act outside their intended permissions. Example: modifying a URL from /user/123 to /user/124 to access another user's data (IDOR).
A02 - Cryptographic Failures	Sensitive data exposed due to weak/absent encryption. Example: passwords stored as MD5 hashes or sensitive data transmitted over HTTP.
A03 - Injection	Attacker-supplied data is interpreted as commands. Example: SQL injection — entering ' OR '1'='1 into a login field to bypass authentication.
A04 - Insecure Design	Fundamental design flaws that cannot be patched — only re-architected. Example: no rate limiting on a password reset endpoint.
A05 - Security Misconfiguration	Default credentials, open S3 buckets, verbose error messages, unnecessary services running. Example: admin panel accessible at /admin with default credentials.
A06 - Vulnerable and	Using libraries with known CVEs. Example: running an old version

Vulnerability	One-Line Description + Example
Outdated Components	of Log4j vulnerable to Log4Shell (CVE-2021-44228).
A07 - Identification and Authentication Failures	Broken auth mechanisms — weak passwords, no MFA, exposed session tokens. Example: session ID visible in URL, never expires.
A08 - Software and Data Integrity Failures	Code or data loaded without integrity verification. Example: CI/CD pipeline using an unverified third-party plugin that injects malicious code.
A09 - Security Logging and Monitoring Failures	Insufficient logging means attacks go undetected. Example: failed logins not logged; no alert for repeated brute-force attempts.
A10 - Server-Side Request Forgery (SSRF)	Server makes requests on behalf of attacker. Example: url=http://169.254.169.254/latest/meta-data/ fetches AWS instance metadata from a cloud server.

Part B — Web Testing - Command Recall Drill (15 Entries)

Cover the "Command / Tool" column. Recall the exact syntax before revealing.

Task	Command / Tool
1. Directory/file brute-force on a web server	<code>gobuster dir -u http://target.com -w /usr/share/wordlists/dirb/common.txt</code>
2. Subdomain enumeration	<code>subfinder -d target.com -o subs.txt</code>
3. Web vulnerability scanner (auto-scan)	<code>nikto -h http://target.com</code>
4. Fuzz a web parameter for hidden values	<code>ffuf -u http://target.com/page?id=FUZZ -w wordlist.txt</code>
5. Test a URL for SQL injection	<code>sqlmap -u "http://target.com/page?id=1" --dbs</code>
6. Send a custom HTTP GET request with curl	<code>curl -X GET http://target.com/api/endpoint -H "Authorization: Bearer token"</code>
7. Intercept and modify HTTP requests	Burp Suite – Configure browser proxy to 127.0.0.1:8080, intercept in Proxy tab
8. Enumerate DNS records for a domain	<code>dig target.com ANY</code> or <code>nslookup -type=ANY target.com</code>
9. Scan all ports with service/version detection	<code>nmap -sV -sC -p- target.com</code>
10. Check HTTP response headers for security misconfigurations	<code>curl -I http://target.com</code> (look for missing

Task	Command / Tool
	headers like CSP, X-Frame-Options, HSTS)
11. Brute-force with a specific file extension	<code>gobuster dir -u http://target.com -w wordlist.txt -x php,html,txt</code>
12. Decode/encode Base64 on command line	<code>echo "dGVzdA==" base64 -d (decode) / echo "test" base64 (encode)</code>
13. Extract metadata from all files in a directory	<code>exiftool -r /path/to/directory</code>
14. Analyze encoded/obfuscated web data visually	CyberChef - gchq.github.io/CyberChef (web-based, chain operations)
15. Passively gather OSINT on a domain	<code>theHarvester -d target.com -b google,linkedin,shodan</code>

Part C — CTF Skills - True or False (10 Questions)

Write *T* or *F* for each statement before checking the answer.

#	Statement	Your Answer	Correct Answer
1	You should always check the file type with <code>file</code> before analyzing a CTF challenge.		TRUE - Never trust the file extension; magic bytes reveal the real format.
2	<code>binwalk</code> can only analyze firmware files.		FALSE - <code>binwalk</code> works on any binary file to detect and extract embedded files and archives.
3	It is legal to test web vulnerabilities on any site without permission.		FALSE - Always obtain written authorization. Unauthorized testing is a federal crime (CFAA).
4	<code>steghide</code> can hide data inside JPEG and BMP images.		TRUE - <code>steghide</code> supports JPEG, BMP, WAV, and AU file formats.
5	The flag format in most CTFs is completely random with no prefix.		FALSE - Most CTFs use predictable formats: <code>CTF{}</code> , <code>flag{}</code> , <code>picoCTF{}</code> , etc. Always <code>grep</code> for these.
6	Frequency analysis is effective against one-time pad (OTP) ciphers.		FALSE - OTP is theoretically unbreakable with a truly random key. Frequency analysis works on classical ciphers (Caesar, Vigenère).

#	Statement	Your Answer	Correct Answer
7	strings only shows ASCII text, not Unicode.		PARTIALLY TRUE - Default strings shows ASCII. Use strings -e l or -e b flag for 16-bit little/big-endian Unicode strings.
8	In a CTF, if stuck for 30+ minutes on one approach, you should try a completely different method.		TRUE - Tunnel vision kills CTF performance. Step back, enumerate more, try a different attack surface.
9	OSINT is only useful for finding people's personal information.		FALSE - OSINT covers domains, IPs, email infrastructure, cloud storage, companies, credentials in breach databases, and more.
10	CyberChef is a command-line only tool.		FALSE - CyberChef is a browser-based GUI tool ("The Cyber Swiss Army Knife"). It can also be run locally.

SECTION 6 - CAREER & CERTIFICATION KNOWLEDGE CHECK

Part A — Certification Path Quiz (10 Questions)

Q1. Which certification is considered the gold standard for offensive security / penetration testing?

1. CompTIA Security+
2. OSCP (Offensive Security Certified Professional)
3. CEH (Certified Ethical Hacker)
4. CySA+

► **Answer: B - OSCP**

OSCP requires a grueling 24-hour hands-on exam on a live penetration testing lab. It is

industry-recognized as proof of real practical skill — not just multiple-choice

knowledge.

Q2. Match each certification to its primary focus area:

Certification	Focus Area
CompTIA Security+	Broad security fundamentals — the entry-level baseline for DoD 8570 compliance
GREM (GIAC)	Malware reverse engineering — analyzing and disassembling malicious code
GCFE (GIAC)	Windows forensics and incident response — evidence collection and analysis
CySA+	Blue team threat analytics — behavioral analysis, log review, SOC operations
eJPT	Entry-level penetration testing — practical, beginner-friendly, fully hands-on
AZ-500	Microsoft Azure cloud security — securing Azure workloads, identity, and data

Q3. What are the three tiers of a Security Operations Center (SOC)?

► Answer

•**Tier 1 - Alert Triage Analyst:** Monitors dashboards, reviews alerts, filters false

positives, escalates confirmed incidents.

•**Tier 2 - Incident Responder:** Investigates escalated incidents, performs deeper

analysis, conducts containment.

•**Tier 3 - Threat Hunter / Senior Analyst:** Proactively hunts for threats, performs

forensics, develops detections and playbooks.

Q4. Name three soft skills cybersecurity employers consistently rank as most important.

► Answer (any three)

•**Communication:** Writing clear incident reports; explaining technical findings to

non-technical leadership.

•**Critical thinking:** Approaching novel problems methodically under pressure.

•**Teamwork / collaboration:** Cyber incidents require cross-team coordination.

•**Continuous learning:** The threat landscape evolves constantly; stagnation is

dangerous.

•**Attention to detail:** Missing one log entry can mean missing an entire intrusion.

Q5. What should a cybersecurity portfolio include to stand out for entry-level roles?

► Answer

•CTF writeups (HackTheBox, TryHackMe, picoCTF) documenting your

methodology and tools

•Home lab projects (e.g., Splunk SIEM setup, Metasploitable practice, custom IDS

rules)

•GitHub repository with scripts, automation tools, or forensic utilities you built

- Documented incident response exercises or DFIR case reports

- Blog posts or LinkedIn articles demonstrating technical depth

Q6. What is the CyberCorps: Scholarship for Service (SFS) program?

► Answer

CyberCorps: SFS is a federal scholarship program funded by the NSF that pays full

tuition, stipends, and professional development costs for cybersecurity students. In

exchange, recipients commit to working for a federal, state, local, or tribal government

agency for the same number of years they received funding. It is one of the best

pathways into government cyber careers.

Q7. What networking events and communities should a cyber professional engage with?

► Answer

- **DEF CON / Black Hat:** World's largest hacking conferences; attend even virtually.

- **BSides events:** Local community-run security conferences — more accessible

than DEF CON.

- **ISACA / (ISC)² local chapters:** Professional networking with practitioners and

hiring managers.

•**Discord communities:** TryHackMe, HackTheBox, TCM Security — active daily

mentorship.

•**LinkedIn:** Follow security researchers, engage with posts, document your learning

publicly.

Q8. What are three common entry-level job titles in cybersecurity?

► Answer

Title	Primary Responsibility
SOC Analyst (Tier 1)	Monitor alerts, triage events, escalate incidents
IT Security Analyst	Vulnerability management, policy enforcement, endpoint security
Junior Penetration Tester	Perform authorized security assessments under senior supervision
Incident Response Analyst	Respond to confirmed security incidents, collect and analyze evidence
Cybersecurity GRC Analyst	Governance, risk, and compliance — audits, frameworks (NIST, ISO 27001)

Q9. What is the MITRE ATT&CK Framework and how is it used in a SOC?

► Answer

MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) is a globally

accessible knowledge base of adversary behaviors based on real-world observations. In

a SOC, analysts use ATT&CK to:

- Map observed attacker behavior to specific techniques (e.g., T1059 - Command and Scripting Interpreter)

- Prioritize detection rules around high-frequency tactics

- Communicate findings using a common language across teams

- Conduct threat hunting based on known threat actor playbooks

Q10. Why is maintaining an active GitHub profile important for a cybersecurity career?

► Answer

GitHub serves as a **living technical portfolio**. Recruiters and hiring managers

regularly review GitHub profiles to assess real skill. A strong GitHub shows:

- Practical coding and scripting ability (Python, Bash, PowerShell)

- Familiarity with version control and collaborative development

- Self-directed learning through documented projects

- Contributions to open-source security tools (extremely valuable)

Even pinning CTF writeup repos, custom Volatility plugins, or YARA rule sets

demonstrates initiative that resumes alone cannot.

Use this section for quick reference during timed review sessions. All answers are organized by section.

Section 1 – Linux Forensics

Part A (Flashcards): See two-column table above. Key commands: `ps aux`, `ss -`

`tu``lnp`, `find / -mtime -1`, `strings`, `stat`, `xxd`, `last`, `crontab -l`.

Part B (Multiple Choice): Q1-B | Q2-C | Q3-C | Q4-B | Q5-B | Q6-C | Q7-B

| Q8-B | Q9-B | Q10-C

Section 2 – Memory Forensics

Part A (Flashcards): See command table above. Key plugins: `windows.pslist`,

`windows.psscan`, `windows.malfind`, `windows.cmdline`, `windows.dumpfiles`,

`linux.bash`.

Part B (Fill in the Blank): 1-psscan | 2-volatile | 3-process injection/hollowing |

4-malfind | 5--EncodedCommand | 6-%TEMP% / %APPDATA% | 7-SHA256 hash

| 8-LiME | 9-services.exe | 10-lsadump

Section 3 – Disk Forensics

Part A (Event IDs): 4624=Successful logon | 4625=Failed logon | 4634=Logoff

| 4648=Explicit credentials | 4688=New process | 4698=Scheduled task created

| 4720=New account | 4732=Added to group | 7045=New service |

1102=AUDIT LOG CLEARED | 4768=Kerberos TGT | 4776=NTLM validation |

4663=Object access | 4656=Handle requested | 4657=Registry modified

Part B (Multiple Choice): Q1-B | Q2-C | Q3-B | Q4-C | Q5-B | Q6-C | Q7-C

| Q8-C

Section 4 – Mobile & Malware Analysis

Part A (Flashcards): See mobile table above. Key items: ADB commands, SQLite

databases, acquisition types (logical < physical < chip-off), exiftool for EXIF/GPS, iOS

backup path.

Part B (Scenarios): See detailed scenario answers embedded in Section 4. Core

principle: **Hash → Document → Analyze → Report.**

Section 5 – Web / Bug Hunting

Part A (OWASP): A01=Broken Access Control | A02=Cryptographic Failures |

A03=Injection | A04=Insecure Design | A05=Security Misconfiguration |

A06=Vulnerable Components | A07=Auth Failures | A08=Integrity Failures |

A09=Logging Failures | A10=SSRF

Part B (Commands): gobuster (dir busting), subfinder (subdomains), nikto (vuln

scan), ffuf (fuzzing), sqlmap (SQL injection), curl (HTTP requests), Burp Suite

(proxy/intercept), nmap (port scan), CyberChef (encoding/decoding), theHarvester

(OSINT).

Part C (True/False): 1-T | 2-F | 3-F | 4-T | 5-F | 6-F | 7-Partial T | 8-T | 9-

F | 10-F

Section 6 – Career & Certification

Cert Ladder: Security+ (entry baseline) → eJPT (entry pen test) → CySA+ (blue

team) → OSCP (advanced offensive) → GREM/GCFE (specialist). Cloud path: AZ-500 for

Azure.

Key Concepts: SOC Tier 1/2/3 roles | CyberCorps SFS (NSF scholarship) | MITRE

ATT&CK (adversary behavior framework) | GitHub = living portfolio | Soft skills:

communication, critical thinking, continuous learning.

PROGRESS TRACKER – BRANDONN'S SCORE LOG

Record your results after each study session. Aim to complete at least 3 attempts per section, spaced 24–48 hours apart. Target 90%+ (Mastery) on all sections before camp evaluation.

Section	Date Attempted	Score (#)	%	Status	Notes / Missed Topics
Section 1 - Linux Forensics <i>30 questions total</i>		/30			
		/30			
		/30			
Section 2 - Memory Forensics <i>30 questions total</i>		/30			
		/30			
		/30			
Section 3 - Disk Forensics <i>23 questions total</i>		/23			
		/23			
		/23			
Section 4 - Mobile & Malware <i>20 questions total</i>		/20			
		/20			
		/20			
Section 5 - Web / Bug Hunting <i>35 questions total</i>		/35			
		/35			
		/35			
Section 6 - Career & Certs <i>10 questions total</i>		/10			
		/10			
		/10			
CUMULATIVE TOTAL		/148			Overall Status

★ Scoring Reference

Score %	Status	Action
90-100%	MASTERY	Move on; revisit briefly before assessment.
70-89%	REVIEW	Re-read missed topics; retry in 24 hours.
Below 70%	RE-STUDY	Return to source material; rebuild from fundamentals.

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